

# DengAI — Predicting Disease Spread

## Metrics and Models

October 2026

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### Metrics

#### The competition metric is MAE

$$\text{MAE} = \frac{1}{n} \sum_{i=1}^n |\hat{y}_i - y_i|$$

The average number of cases we are off by, per week.

- One number over all **416 test weeks** (260 San Juan + 156 Iquitos)
- Lower is better; a perfect model scores 0
- **San Juan dominates**: 63% of the weeks and about four times the cases

#### The target is skewed

San Juan: **mean 34, median 19**, 73% of weeks below the mean. Skewness **4.5** (San Juan) and **4.0** (Iquitos).

Mean cases per week are 34.2 in San Juan and 7.6 in Iquitos, but the median week has only 19 and 5; the maxima are 461 and 116. In both cities the worst 10% of weeks hold 43% of all cases.

Skewness is a number for this lopsidedness: 0 means symmetric, like a bell curve; above 1 already counts as strongly skewed. Ours is 4.5 and 4.0.

Everyday analogy: income. A few billionaires raise the *average* income, but the *median* still describes the typical person. For us this means that the rare outbreak weeks would dominate any metric that punishes big errors heavily.

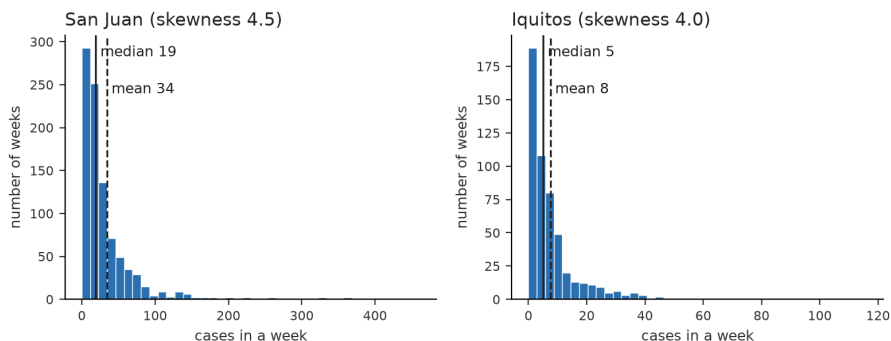


Figure 1: Weekly dengue cases: most weeks are quiet, a few outbreak weeks form a long tail

## Why MAE and not RMSE

- RMSE squares errors: one 461-case week counts like **~500 ordinary weeks off by one**
- RMSE is dragged into fitting a few outbreaks; **MAE weights every week equally**
- MAE is minimised by the **median**, RMSE by the **mean** — for skewed counts the median is the robust target
- Consequence: good models here are *conservative*

Under RMSE the model concentrates on the handful of epidemic weeks and pays little attention to the hundreds of ordinary ones. MAE treats a miss of 10 in a quiet week exactly like a miss of 10 in an outbreak week.

Statistically, the prediction that minimises expected absolute error is the conditional median; squared error is minimised by the conditional mean, which for a right-skewed distribution sits far above the typical week. This also explains a behaviour we see later: the models that do well on this metric predict the quiet-week level and rarely chase peaks.

## Models

### The pipeline has four branches

`make_city_pipeline` builds one of four shapes, because model families need different preprocessing:

| Branch            | Steps                                           | Models                    |
|-------------------|-------------------------------------------------|---------------------------|
| <code>tree</code> | impute → history → cyclical →<br>select → model | CatBoost,<br>RandomForest |

| Branch             | Steps                                     | Models                       |
|--------------------|-------------------------------------------|------------------------------|
| <b>dense</b>       | tree steps + <b>impute</b> + <b>scale</b> | Ridge, Lasso                 |
| <b>time_series</b> | impute → history → date + scaled weather  | Prophet, SARIMAX             |
| <b>seasonal</b>    | calendar columns only                     | Seasonal median, Shape×level |

## Why four branches

**dense** — the lag/rolling warm-up leaves **NaNs** in the first 26 weeks

- Tree models handle NaN natively; RidgeCV raises `ValueError`
- So linear models get `SimpleImputer(median)` + `StandardScaler`

**seasonal** — the calendar baselines need `year` and `weekofyear`, which the tree branch drops

- They **never see the weather**: the baseline any weather model must beat

Scaling matters for the linear models too: Ridge and Lasso penalise coefficient sizes and are not scale-invariant, so unscaled features would be penalised unevenly.

The tree branch drops `city`, `week_start_date` and `weekofyear` before the model. Rather than weaken that branch, the two calendar models get their own: a `ColumnTransformer` passing through `year` and `weekofyear` and nothing else.

## The nine models

| Model           | Family               | Core assumption                                     |
|-----------------|----------------------|-----------------------------------------------------|
| CatBoost        | gradient boosting    | non-linear effects of lagged weather                |
| XGBoost         | gradient boosting    | as CatBoost, on RFE-selected features               |
| RandomForest    | bagged trees         | same, without boosting                              |
| Ridge           | linear + L2          | smooth linear response, all features shrunk         |
| Lasso           | linear + L1          | linear, but most features are irrelevant            |
| Prophet         | additive time series | trend + annual seasonality + regressors             |
| SARIMAX         | ARIMA + exog         | autocorrelated errors + Fourier seasonality         |
| Seasonal median | calendar only        | this week behaves like the same week in past years  |
| Shape × level   | calendar only        | seasonal <i>shape</i> × typical annual <i>total</i> |

## Results — holdout MAE

| Model           | San Juan     | Iquitos |
|-----------------|--------------|---------|
| <b>CatBoost</b> | <b>15.02</b> | 6.27    |
| RandomForest    | 15.16        | 8.39    |
| XGBoost + RFE   | 17.21        | 3.54    |

| Model                                  | San Juan | Iquitos     |
|----------------------------------------|----------|-------------|
| <b>Shape <math>\times</math> level</b> | 20.72    | <b>3.39</b> |
| Lasso                                  | 21.16    | 4.50        |
| Prophet                                | 21.78    | 7.16        |
| Seasonal median                        | 23.49    | 4.41        |
| Ridge                                  | 24.46    | 5.48        |
| SARIMAX                                | 24.56    | 4.61        |

Different winners per city — the submission uses **CatBoost for San Juan, SARIMAX for Iquitos**.

XGBoost comes from a separate notebook (`experiments/xgboost_feature_selection.ipynb`) that compared feature-selection strategies on chronological CV (3 folds of 52 weeks). CV MAE: all 127 features 17.8 (San Juan) / 7.2 (Iquitos); the top 100 by XGBoost importance 17.0 / 7.1; recursive elimination with Ridge 14.5 with 50 features / 6.7 with 10 features. On the same 52-week holdout the RFE model scores 17.21 and 3.54: third in San Juan, and the best weather-based model in Iquitos, where ten long-window features (8–26-week temperature means, 26-week rainfall and humidity, the annual sine and cosine) suffice. A linear model chose features better than the trees’ own importances did.

## What drives the CatBoost predictions?

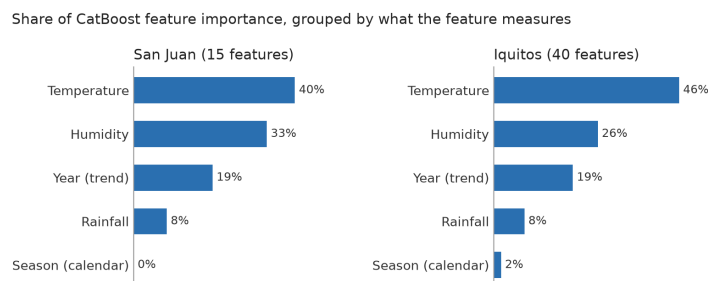


Figure 2: CatBoost feature importance by feature family

- **Temperature and humidity: about 70%** of importance in both cities
- Top features are **8–26-week averages**, not this week’s weather
- **year: about 19%** — levels drift between years for non-weather reasons

Mosquitoes breed and bite more when it is warm and humid. Weather acts with a delay: the mosquito population has to build up first, which is why the long windows win.

Rainfall matters less than expected (about 8%): temperature and humidity already capture most of what the rain does.

Importance tells us what the model *uses*, not what *causes* dengue.

## Why weather is not the whole story

- Climate predicts **when** cases rise within a year, not **how many**
- In Iquitos the calendar-only **Shape × level** (3.39) beats every weather model
- Epidemic size depends on the circulating **serotype** and population immunity — neither is in the data

Feature importance and the results table point the same way. A model that predicts the shape of the year from the calendar and refuses to guess the level from weather is not naive; it is honest about what the data supports.

## Summary

- **Metrics** — MAE, because the target is skewed (4.5): a few outbreak weeks would dominate RMSE
- **Models** — nine models in four pipeline branches; winners differ by city
- **Feature importance** — months of temperature and humidity; weather says *when*, not *how many*

A side experiment with XGBoost confirms that feature selection matters: 50 features beat 127 in San Juan, and 10 are enough in Iquitos.